Received: 10 January 2023 | Accepted: 14 October 2023

DOI: 10.1111/mice.13117

RESEARCH ARTICLE

COMPUTER-AIDED CIVIL AND INFRASTRUCTURE ENGINEERING WILEY

Deep learning-based automatic classification of three-level surface information in bridge inspection

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Funding information
National Natural Science Foundation of China, Grant/Award Numbers: 52122801, 11925206, 51978609, U22A20254; Science Fund for Distinguished Young Scholars of Zhejiang Province, Grant/Award Number: LR20E080003

Abstract

Bridge inspection ensures that in-service bridges are managed and maintained in conformity. To enhance the accuracy and efficiency of bridge inspection, an automatic hierarchical model is proposed, which enables the classification and correlation of bridge surface images at three levels, namely, at the structure, component, and defect type level. Thus, the impact of both the defect types and the affected components on bridge safety can be simultaneously considered. The proposed model uses a group of sub-models instead of the common flat network to realize the multiple tasks, which is advantageous in accuracy, training simplicity, and scalability. The classification accuracy of the hierarchical model in three levels has reached 96%, 92%, and 81%. Results demonstrate the effectiveness of the proposed method in the classification of multi-scale targets. This study may provide a new strategy for developing a systematic and easily adaptable detection framework for practical bridge engineering.

1 | INTRODUCTION

As a result of material aging, increasing traffic loads, and accidents, bridges worldwide have entered a phase of gradual deterioration. According to a survey conducted by the American Society of Civil Engineers (ASCE, 2017), there are currently more than 600,000 bridges in the United States, of which at least 42% have been in service for over 50 years. In China, the percentage of bridges aged 30 years or more is projected to rise from 16.1% in 2014 to 62.7% by 2044. This impending increase in bridge aging poses a growing concern, amplifying the risk of bridge failures. Therefore, timely and accurate structural health monitoring (SHM) of bridges is crucial for ensuring their long-term safety and serviceability. However, conventional manual inspection methods are often time-consuming, labor-intensive, and prone to low accuracy and oversight possibilities (Chun et al., 2021; Dong & Catbas, 2021).

To address these challenges, artificial intelligence (AI) techniques, such as machine learning (ML) and computer vision (CV), have been introduced to enable autonomous SHM of various structures and infrastructure, including bridges (Mirzazade et al., 2021; Zoubir et al., 2022), pipelines (Y. Li et al., 2023; Ma et al., 2022), tunnel structures (Marasco et al., 2022; Rosso et al., 2023), and pavements (Rodriguez-Lozano et al., 2023; L. Zhao et al., 2022), among others (Ye et al., 2023). These AI methods provide promising solutions by leveraging their capabilities in data analysis, pattern recognition, and automated decision-making. By integrating ML and CV techniques into bridge SHM, it is possible to enhance the efficiency, accuracy, and effectiveness of monitoring

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processes, thereby mitigating risks and improving the overall safety of bridge infrastructure.

The SHM can be classified into two main categories based on data type (Rafiei & Adeli, 2017b): vision-based (Sirca et al., 2018) and vibration-based (Amezquita-Sanchez & Adeli, 2015; Pezeshki et al., 2023; Y. Zhang et al., 2022). Similarly, the application of ML techniques in SHM follows a comparable division. Conventional methods encounter significant challenges in handling high-dimensional and unstructured image data, whereas ML has demonstrated notable advantages in the domain of image recognition. Consequently, numerous ML methods have been proposed for vision-based SHM tasks, including damage detection (Lin et al., 2022), nondestructive evaluation (Xiaofeng Li et al., 2022), deformation or displacement monitoring (J. Zhao et al., 2022; Zhuge et al., 2022), and 3D reconstruction (Wei et al., 2022). It is worth mentioning that the integration of advanced algorithms, such as ML, in vibration-based SHM is also an active area of research. This line of research focuses on aspects such as response information (Corbally & Malekjafarian, 2023; Oh et al., 2017; H. Zhang et al., 2023), modal identification (Z. Li et al., 2017; Perez-Ramirez et al., 2016), data recovery (Jiang et al., 2022), and response prediction (Perez-Ramirez et al., 2019), among others (Amezquita-Sanchez et al., 2018; Soleimani-Babakamali et al., 2022).

In the aforementioned ML-based SHM research, damage identification has received much attention and related algorithms and methods have developed rapidly. One of the typical domains is crack detection using a convolutional neural network (CNN). Early models like CrackNet-V were limited in depth and simplicity (L. Zhang et al., 2016; A. Zhang et al., 2017), resulting in suboptimal performance. To address this, researchers proposed deeper CNNs. For instance, Cha et al. (2017) introduced an eight-layer CNN for concrete crack detection achieving 98% accuracy, while Xu et al. (2019) developed a 28-layer CNN for bridge crack identification with 96.37% accuracy, even without pre-training. However, bridges are susceptible to various defects beyond cracks, such as rebar exposure, spalling concrete, and pitted surfaces. While some CNNs excel at crack identification, their precision and efficiency decrease when applied to multiple defect recognition. Visual Geometry Group 16 (VGG16, Simonyan & Zisserman, 2015), a deep CNN, demonstrates promise in recognizing multiple bridge defects. Wu et al. (2019) pruned and utilized VGG16 to classify corrosion and crack defects in bridge structures, achieving accuracy rates of 93.6% and 98.5%, respectively. Novel networks like Faster Region-CNN (Faster RCNN) and You Only Look Once (YOLO) series (Chun et al., 2023; Z. Zhou et al., 2022) have emerged for object detection, while Mask RCNN and U-Net (Yamaguchi & Mizutani, 2023) are employed for segmentation. Recent researchers proposed EfficientNet and incorporated attention mechanisms (Chen & He, 2022; Y. Pan & Zhang, 2022; L. Zhang et al., 2023), separable convolution (Zhu et al., 2023; Zou et al., 2022), deformable convolution (Lei et al., 2023), atrous convolution (Siriborvornratanakul, 2023), and other strategies (Zheng et al., 2022) to further enhance model performance.

Despite significant advancements in defect detection algorithms, most existing studies focus on surface defects in building materials like concrete or steel. However, bridges, as complete structures, exhibit a more complex and specific range of defects. Chun et al. (2022) identified 27 types of defects based on "Bridge periodic inspection guidelines" (2019) in Japan, while the Chinese bridge guidelines list over 70 types of defects across different components. To accurately diagnose bridge conditions, the impact of all defects must be considered, including easily overlooked defects on common components (e.g., water seepage at beam joints) and defects specific to bridge components (e.g., rubber bearing deformations). Research on these defects remains limited, with separate models for different defect types. In this case, it would require multiple iterations of recognition by different models to determine the presence of any defects in the image, which is inefficient. Incremental learning is a potential solution for expanding or modifying defect recognition categories in a model. However, it is considered challenging due to factors such as hyperparameter sensitivity (Goodfellow et al., 2013). Although there are methods available to facilitate incremental learning, the process of retraining and tuning are cumbersome in general and may destabilize the model. Y. Liu et al. (2018) developed visual tree-CNN (VT-CNN) for addressing confusion in categorization, while Roy et al. (2020) proposed Tree-CNN to address catastrophic forgetting. Their network incorporates the idea of the hierarchical structure, but it is used to divide coarse-grained information (Dogs) into fine-grained information (Labrador). That is, once the final network identification result is obtained, the information in the previous levels will be redundant.

Moreover, the bridge surface (BS) images contain information on not only the defect types but also the damaged components, which are often overlooked but crucial. This is because even identical defects can have varying impacts on the bridge's health depending on the structural components involved. Chun et al. (2022) proposed a method that can innovatively output integrated information on defects and members in the form of subtitles, showing great application potential (Kunlamai et al., 2023). Although it includes various types of members and defects, it does not consider the issue of future incremental learning.

In the field of SHM for bridges, there remains a scarcity of systematic and easily updatable detection frameworks

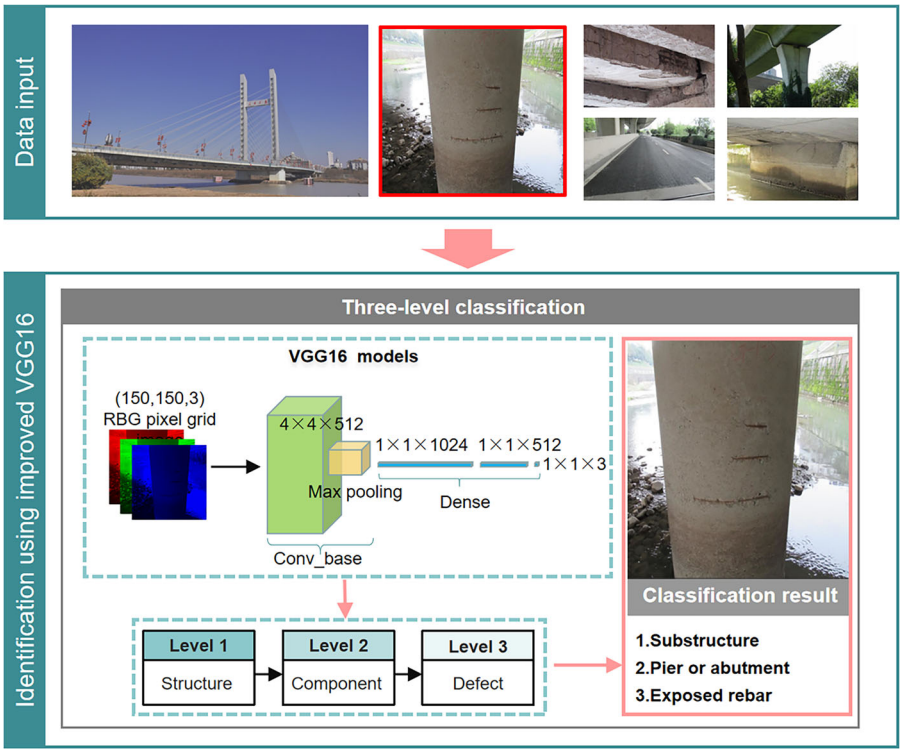


FIGURE 1 Framework of the automatic three-level classification methodology.

and methods. Hence, this study proposes a hierarchical classification and correlation method for BS information at three levels: structural types, components (23 in total), and defects (75 in total). The three-level classification method not only effectively reduces the confusion between defects in different bridge components thus improving the accuracy but also enhances the training flexibility and scalability of the model by dividing the task. Additionally, the simultaneous recognition of the defects and the related components helps in providing a comprehensive understanding of the operation status of the bridge.

2 | METHODOLOGY

2.1 | Overview

Herein, a hierarchical classification model with deep CNNs for bridge defect and component classification is proposed. The overall workflow of the model is shown in Figure 1. Taking an image of exposed rebar as an example, the result identified by the three-level classification model (i.e., structure level, component level, defect level) will be "1. substructure, 2. pier or abutment, 3. exposed rebar." As a hierarchical CNN, the proposed model has the following advantages. In terms of methodology, the hierarchically structured model has been demonstrated to exhibit superior performance in scalability, training simplicity, and training flexibility. Additionally, the proposed classification model demonstrates a high level of accuracy in categorizing over 70 distinct defects, achieving the defect coverage outlined in bridge inspection guidelines. The following sections will provide comprehensive insights into the model's design and its associated advantages.

2.2 | A three-level classification model for BS information

2.2.1 | Hierarchical structured model

Classifying images level by level in a tree-structured workflow is a simple idea but has been demonstrated as an effective approach for traditional linear classification models (B. Liu et al., 2013). There are also some studies on tree-structured CNNs or say hierarchical CNNs (Yan et al., 2015) as introduced in Section 1, further demonstrating the advantage of tree-structured networks in large-scale image recognition and incremental learning capability. However, most of these studies focus on applying tree-structured networks to fine-grained image recognition, which means that there is essentially only one task. In other words, when the output (Labrador) at the third level is obtained, the recognition results of the former two levels (dog and animal) are no longer meaningful and, as a result, can be questioned as laborious.

Based on a similar idea, a group of sub-models was used instead of a single model to accomplish the multitask of BS information recognition by classifying the types of structures, components, and defects corporately. However, although the whole model appears to be a branching structure like a tree, it is not in the traditional sense of top–down decomposition. Instead, it is more like a topological model built on the relationships between categories. The order of the hierarchy incorporates more prior knowledge rather than starting from coarse-grained to fine-grained. Therefore, the proposed model maintains the relative efficiency for keeping different levels responsible for diverse tasks. The proposed model will be further elaborated in the following section with detailed explanations.

2.2.2 | The establishment of a three-level classification model

With over 70 types of BS defects and the potential for their occurrence on various components, the comprehensive identification of BS information poses a relatively large-scale classification problem. The increase in the number of categories leads to a rapid escalation in the difficulty of image recognition due to factors such as computational complexity, category imbalance, and inter-class similarity. To address this problem, 17 sub-models are established to disassemble the task (Figure 2).

The structure types, namely, "bridge floor," "superstructure," and "substructure," will be identified from the images using M_1 as the initial step. Then, the component types encompassing 23 categories such as the traffic sign, deck, sidewalk, and so forth, will be figured out by M_{2_1} to M_{2_3}. It is worth mentioning that, the 23 types in the second level, actually include one type of defect, that is, surface ponding, which belongs to the drainage system. This setting is to prevent confusion caused by the feature similarity between the surface ponding and the deck (one type of bridge component). Therefore, there are only 22 real components in total. Additionally, the images of beam bridges in the second level will be further divided into beam members and transverse connections by an extra model, before entering the third-level identification. These two points are specifically stated here to avoid misunderstanding. The other 13 sub-models ($M_{3_1}, M_{3_2} \ldots M_{3_13}$) are used to classify the 74 categories of defects. Since not all components are susceptible to damage, for example, traffic signs, there are fewer sub-models in Level 3 than the component numbers. Since the defect of surface ponding is classified at Level 2, the model can identify a total of 75 bridge defects (Appendix).

Under the three-level structure, each sub-model is supposed to classify only 4–11 categories of defects, resulting in a reduction in the number of training parameters for each sub-model. Meanwhile, the network can be tuned more specifically according to the training performance of each sub-model, enhancing the flexibility of training. Since the sub-models can be trained in parallel, the training time is greatly reduced. Furthermore, as the tasks of different sub-models at the same level are similar, transfer learning can be effectively employed among the sub-models to share bottom-level parameters and expedite training.

Due to the three-level model, defects belonging to different components but with similar characteristics can be triaged at the component level, reducing the interference in the process of defect identification. For example, exposed rebar on beams and corrosion of steel pads have similar morphology and are easily confused when identified together (see Section 4). However, according to the hierarchical model, these two types of defects are assumed to be located on different components and identified with different sub-models, which prevents confusion in identification with the same model.

The hierarchy order is deliberately devised in this study. The reason why we identify the types of components first and defects second is that the same type of defects may have different geometric characteristics when located in different components. For instance, cracks in sidewalk tiles are often reticulated and intersect with the tile pattern, while bridge deck cracks are linear and often involve water seepage and efflorescence. The intra-class differences between defects are larger than that between components. The large intra-class differences would increase the difficulty of classification, so the images are divided into different components first thus solving the problem.

In general, the proposed three-level model is designed to reduce the difficulty of large-scale classification and accomplish the task of simultaneous identification of defects and components. At the same time, it is found that such a structure introduces relationships between categories, thus improving accuracy and interpretability.

2.3 | An improved VGG16-based network for BS information classification

To address 17 distinct sub-tasks, the three-level classification model necessitated the utilization of multiple sub-models. Hence, the classic VGG16 is adopted since it is relatively lightweight and easily trainable. The hierarchical arrangement effectively partitions the overall task and incorporates knowledge of class relationships, resulting in a reduction in classification difficulty. As a result, the network based on VGG has already exhibited satisfactory performance in tackling the task discussed in this research. The adoption of a three-level structure allows for

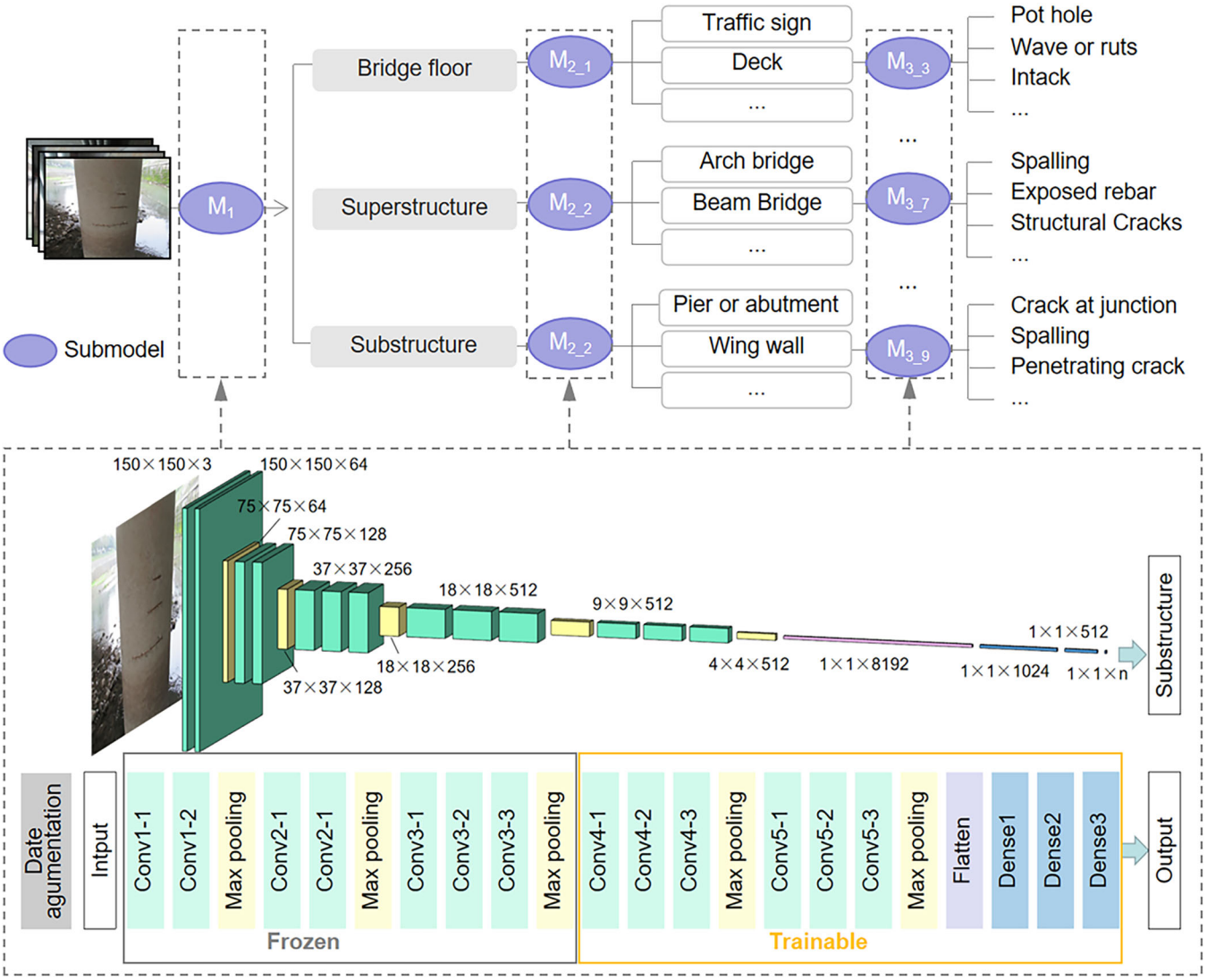


FIGURE 2 The three-level classification model based on improved VGG.

the assignment of each sub-task to a specific network, tailored to its inherent complexity. Therefore, each sub-model here can be replaced with other more advanced networks in the future.

The original VGG16 model (the initial model in Figure 3) consists of five stacked convolutional blocks and three dense layers (Figure 2). VGG adopts the strategy of repeated small 3×3 convolutional kernels, which serves to enhance nonlinear mapping capabilities while reducing the number of parameters involved. Following the processing by the convolutional blocks, the multidimensional input undergoes flattening via the flattened layer, resulting in a one-dimensional vector. This vector is subsequently fed into the subsequent three dense layers for further fully connected processing. The final dense layer is activated using a softmax function, producing probability values for each node. Through this process, VGG16 identifies the most probable node, allowing for multi-classification predictions to be obtained.

To improve the performance of VGG, a parameter-based transfer learning approach was implemented. It is noticed that the literature on transfer learning employs various terminologies, leading to inconsistencies. For instance, terms such as transfer learning and domain adaptation (Wang & Deng, 2018) are used interchangeably, despite having slightly different meanings. Therefore, to ensure clarity and precision, it is worth noting that the expressions used herein refer to the definitions and interpretations presented by Weiss et al. (2016) and S. J. Pan and Yang (2010).

The implementation is outlined as follows. Initially, the convolutional base's weights are directly transferred from the pre-trained model, while the dense layers are replaced with three newly defined dense layers. The first dense layer is configured with a reduced number of neurons,

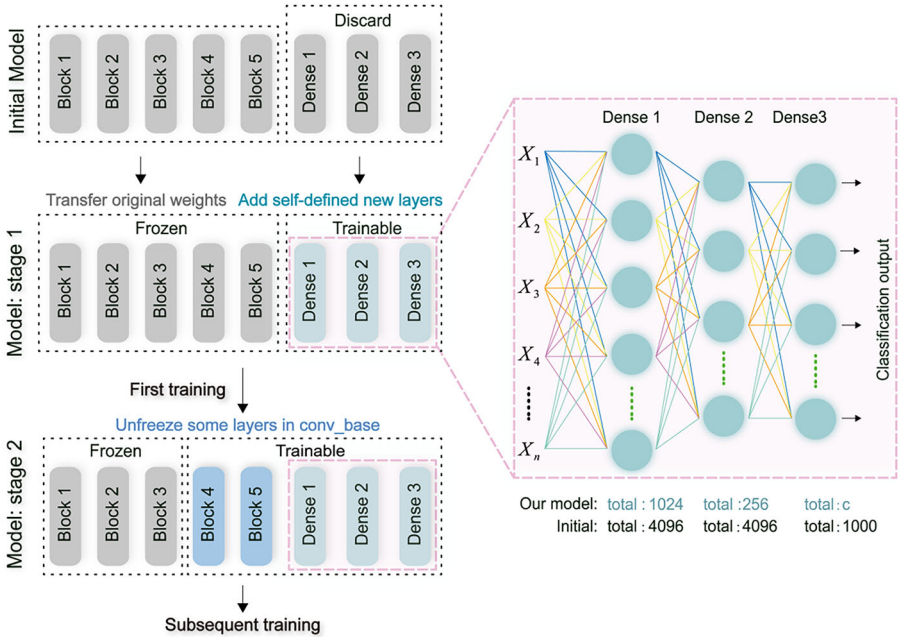


FIGURE 3 Fine-tuning strategy and modification of improved VGG16.

specifically 1024, while the second layer is chosen to have either 256 or 512 neurons (Stage 1 in Figure 3). During the initial training phase, the convolutional base is frozen and only the new dense layers are trained to prevent drastic changes in the weights of the convolutional base. These changes, caused by the dense layers with randomly initialized weights, can destroy the originally learned features. Generally, the lower convolutional layers (e.g., layers within Blocks 1 and 2) tend to extract lower-level features such as edges, which have broad applicability across multiple tasks. In contrast, the higher layers possess the ability to extract more complex and semantically significant features that are task-specific (Aggarwal, 2018). So, in the subsequent training, the higher layers in Blocks 4 and 5 are set to be trainable alongside the dense layers, while the remaining blocks are kept frozen. This configuration empowers the model to acquire new knowledge and perform better.

2.4 | Model compilation

After determining the network, the model compilation needs to be configured, which mainly depends on the choice of three parameters: loss function, optimizer, and monitoring metrics.

The loss function is used by deep learning (DL) models to measure the distance value between the predicted value and the true value. In the training, the loss reduction is made as the target, and the distance value is used as the basis for adjusting the weights. Since this paper performs the multi-classification of images containing defects, the cross-entropy function is adopted, and its formula can be written as follows:

$$L = \frac{1}{n}\sum_{i=1}^{n} L_i = -\frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{m} y_{ic} \ln p_{ic} \tag{1}$$

where n is the number of samples, m is the number of categories, y_{ic} is the expected output, and p_{ic} is the actual output of the neuron.

Optimizer is a mechanism to update the network weights based on the loss function and training data, while gradient descent is one of the most widely used algorithms in neural networks today. Three common gradient descent algorithms, Stochastic Gradient Descent (SGD), Root Mean Square prop (RMSprop), and Adam are tried and compared. Other things being equal, the model performs best using the RMSprop algorithm. So, the RMSprop algorithm is adopted, and the decay rate is introduced to regulate the model learning rate and speed up the training convergence. Its formula can be written as follows:

$$\omega_{t+1} = \beta\omega_t + (1-\beta)g_t^2 \tag{2}$$

$$\Delta x = -\eta \frac{g_t}{\sqrt{\omega_{t+1}} + \varepsilon} \tag{3}$$

where ω_t is the moving average of the squared gradient, g_t is the sample average gradient, β is the decay rate, Δx is the learning rate change value, and η is the initial learning rate.

Accuracy is the ratio of the number of correctly classified images to the total number, which can better reflect

TABLE 1 Compilation settings for all sub-models.

Compilation content	Settings
Loss function	Cross-entropy function
Optimizer	RMSprop
Monitoring metrics	Accuracy

Abbreviation: RMSprop, root mean square prop.

the global sample prediction. It is defined as follows:

$$Accuracy = n_correct/n_total = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

Precision refers to the proportion of positive samples contained in all samples predicted to be positive, and its formula is:

$$P = \frac{TP}{TP + FP} \quad (5)$$

where *TP* (true positive) indicates the number of judgments that are correct in the sample judged to be positive; *FP* (false positive) indicates the number of incorrect judgments in samples judged to be positive; *TN* (true negative) indicates the number of judgments that are correct in samples judged to be negative; *FN* (false negative) indicates the number of incorrect judgments in the negative samples.

The model compilation of the improved VGG16-based network is summarized as shown in Table 1. For more related content regarding model compilation, refer to Aggarwal (2018), LeCun et al. (2015), and Géron (2022).

3 | DATASETS CONSTRUCTION

The establishment of datasets is crucial since ML algorithms acquire the patterns and internal rules of target objects from a mass of data (Sun et al., 2017). This section will introduce the establishment of image datasets for BS information's three-level classification.

3.1 | Data source: From actual bridge inspections

Abundant and representative raw data are the precondition of reliable datasets. So more than 50,000 photographs from on-site inspections are collected as raw data considering three aspects: the quantity of raw data, the identification tasks, and the sample diversity and authenticity: (1) A huge number of images are needed since the identification targets add up to nearly 100 types, and each type requires quantities of images. (2) Each image is supposed to contain three levels of BS information simultaneously, including structural types, components, and defects. This makes it possible to identify multiscale targets through the same image. (3) To ensure sample diversity and authenticity, the images in this study are taken from on-site inspections of hundreds of in-service bridges in Zhejiang Province. Moreover, there is various interference information in the images such as plants, shading, scratches, and debris, which are beneficial for improving the robustness of the model. Additionally, data augmentation is used here to further ensure the diversity of data.

3.2 | Datasets for three-level classification of BS information

The datasets for the three-level classification of BS information are established as follows. The overall dataset contains 32,431 samples, all in jpeg format. For the first-level classification dataset, the training set contains 4535 samples (2368 for the bridge floor, 723 for the superstructure, and 1444 for the substructure), and the validation (VLD) set contains 1170 samples (595 for the bridge floor system, 180 for the superstructure, and 395 for the substructure). The statistics of the second-level and the third-level datasets are shown in the Appendix.

To minimize manual labor, the labeling process (Figure 4) herein has five main steps. (1) Manual classification: Only about 500 to 2000 images still need to be manually identified and labeled by categories to form a preliminary dataset. The original labels cover 75 types of defects as mentioned in Section 2.2.1. Based on the experience of manual classification, some adjustments are made subsequently to the labeling of defects. First, some defects with similar features are combined. For instance, "structural crack" and "surface crack" are both labeled as "crack." Second, multiple labels, like "exposed rebar and spalling," are added to cope with the simultaneous appearance of two defects. (2) Model training: With a small and easily constructed initial dataset, model training and optimization can be conducted efficiently, leading to the rapid attainment of a relatively optimal model.(3) Model recognition: Although the early model may have relatively low accuracy due to limited training data, it still possesses classification capabilities. Thus, the model can be used to label the remaining unclassified images from the gallery automatically. (4) Manual modification: Due to the possibility of incorrect labels based on model predictions, manual verification and modification are necessary. The images with revised labels will then be added to expand the dataset. (5) Model retraining: The expanded datasets will be used to retrain the network to update the weights and optimize the model. The (3) to (5) is supposed to be cycled until the model classification ability has reached the expectation.

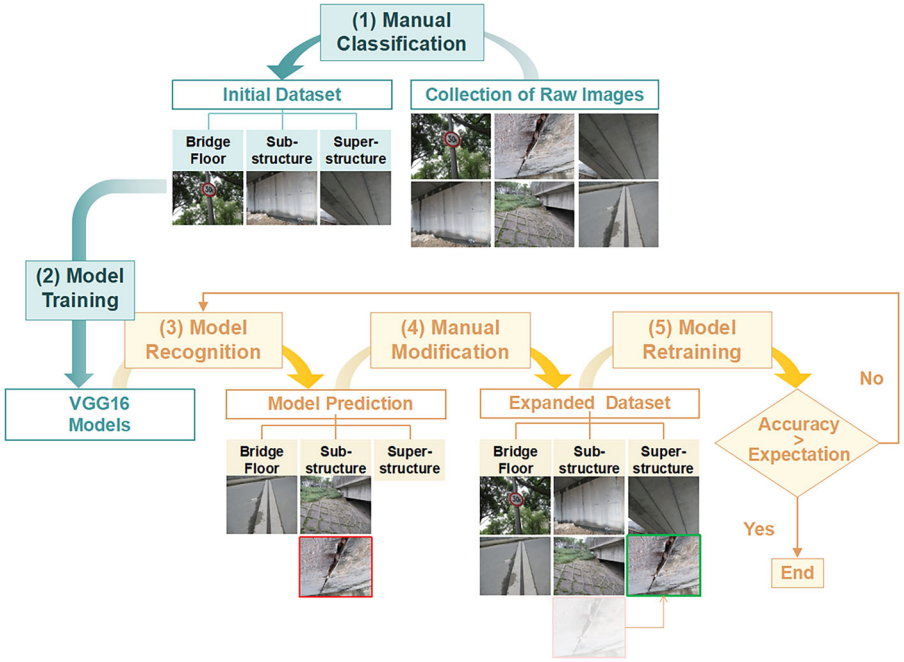


FIGURE 4 Labeling process.

TABLE 2 The training and validation results of different networks.

Case	Network	Unfrozen blocks	Added layers	Neuron numbers	Accuracy		Loss	
					Training	VLD	Training	VLD
Case 1		—	—	4096, 4096	50.0%	49.0%	0.982	0.978
Case 2	Original	B5	—	4096, 4096	69.2%	69.4%	0.742	0.756
Case 3	Fine-tuned	B4 and B5	—	4096, 4096	96.6%	95.5%	0.201	0.234
Case 4	Fine-tuned	B4 and B5	—	512, 512	94.5%	94.7%	0.242	0.281
Case 5	Fine-tuned	B4 and B5	—	256, 256	97.1%	95.1%	0.159	0.231
Case 6	Fine-tuned	B4 and B5	—	1024, 256	98.1%	96.0%	0.050	0.226

The semi-automated labeling process not only ensures the correctness of labels by step (4) but also reduces the large amount of manual labeling work. In this process, the three tasks of the dataset construction, model training, and classification are not separated as usual but coupled with each other and updated jointly. This may facilitate the automation of classification problems and rapidly improve the accuracy and stability of classification models.

4 | EXPERIMENTS

4.1 | Training and hyperparameter optimizations

Although this study uses 17 sub-models to enhance the flexibility of single-network training, the overall training workload is heavy. However, due to the similarity of tasks amongst sub-models, particularly within the same level, transfer learning is widely used herein to improve training efficiency. Among these tasks, the first-level classification of structure types is considered the easiest. Therefore, M_1 is trained first based on the pre-trained VGG on ImageNet and serves as a basis for subsequent sub-models. Considering the training of M_1 as an example, Table 2 demonstrates the process of adjusting the relevant hyperparameters in the transfer learning process. The overall results indicate that the improved VGG (Case 6) performs best and achieves the highest accuracy with a 45.0% improvement, compared with the original VGG (Case 1), showing the effectiveness of the transfer learning strategy adopted. When transfer learning was performed on other sub-models, only the pre-trained weights were changed, but the overall configuration remained similar to that of Case 6.

Extensive investigation of hyperparameter configurations is conducted during the training process to further optimize the models. Since the dataset sizes and feature information differ among various sub-models, the optimal hyperparameter settings for sub-models are presented as ranges as summarised in Table 3.

TABLE 3 Hyperparameters after optimization.

Hyperparameter	Value
Epoch	[100, 150, 200, 250]
Initial learning rate	$1 \times 10^{-4} \sim 4 \times 10^{-6}$
Decay	[0.8, 0.9]
Patience	5 (epochs)
Batch	128/256

4.2 | Baseline

Although advanced models have shown excellent multi-classification capabilities, such as VGG achieving high accuracy on the 1000-class ImageNet task, the use of them cannot be directly assumed to be successful in recognizing multiple BS information. This limitation arises from the presence of noises and interferences, such as messy backgrounds of inspection images. Moreover, the defects are small and of various shapes, creating many challenges to multi-defect detection in practical scenarios.

Currently, most defect recognition methods adopt a flat structure workflow (Figure 5a). However, as discussed in Section 2.2, using a three-level structure for classification presents certain advantages for dealing with actual defects in bridge engineering. This section presents a case study and comparison of flat and multilevel models.

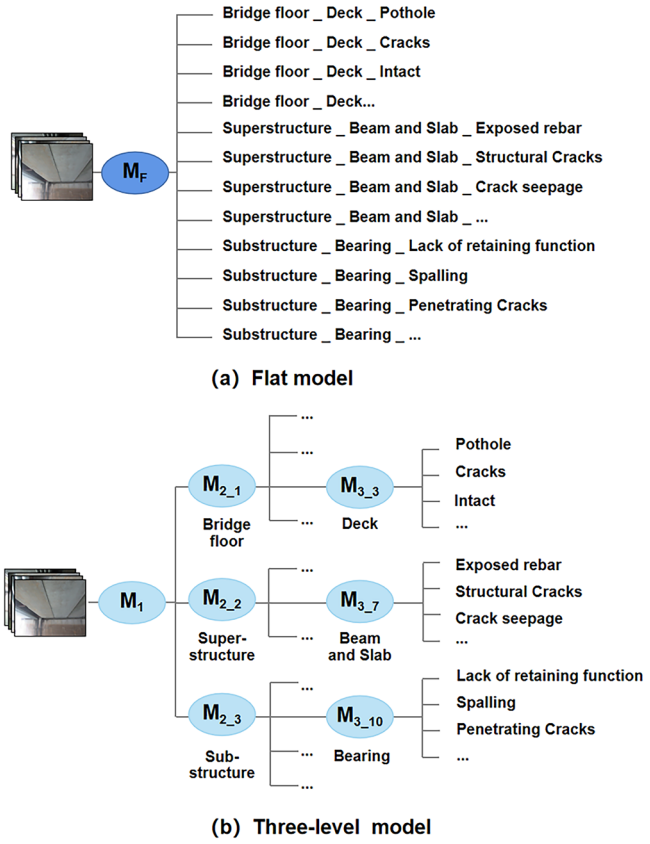


FIGURE 5 Flat model and three-level model.

TABLE 4 The performance of the flat model and three-level model.

Model		Accuracy		Loss	
		Train	VLD	Train	VLD
Flat model		45.92%	34.18%	1.86	2.39
Three-level model	Deck	75.85%	71.02%	0.41	0.48
	Beams and slabs	80.88%	60.62%	0.60	0.91
	Bearing	92.91%	89.92%	0.24	0.37
	Average	83.21%	73.85%	0.42	0.59

Since this study involves 75 types of defects in different components, direct training of an equivalent flat model with 75 classes would require considerable training time. Therefore, a scaled-down flat model is used (Figure 5a), which includes 24 types of defects related to three types of components in the deck, beam and slab, and bearing, so that these selected types can be representative to a certain extent. By comparing the results of the scaled-down model and the corresponding three-level model (Figure 5b) on these 24 defect classification tasks, the difference between the two models can be more clearly understood.

The models are trained using the same dataset, which contains 583 images (the training set and validation set are split at a 4:1 ratio). The results (Table 4) show that the training accuracy of the flat model remained stable at approximately 45.92%, with a validation accuracy of 34.18%. Around 40 epochs, the validation accuracy starts to exhibit a decreasing trend (Figure 6a), suggesting that the network is experiencing overfitting. In the three-level model, the training accuracy (validation accuracy) of the sub-models in Level 3 is 92.91% (89.92%) for bearing defects (Figure 6c), 80.88% (60.62%) for deck defects, and 75.85% (71.02%) for defects on beams and slabs. The average validation accuracy is 74.89%. This outcome is expected because task division reduces the complexity of classification and should improve the performance of single sub-models. However, the accurate classification of images in the three-level model also depends on the first two levels. Figure 6d depicts the training process of the sub-model of substructure, which is typical for the first two level sub-models. The overall accuracy (only the images that are identified correctly in all three levels are considered as correct cases) of the three-level model is tested to be 65.53%, which demonstrates a twofold improvement compared to the flat model.

The performance of two kinds of models trained on larger dataset sizes is further investigated and compared. When the dataset expands to 1513, the accuracy of the flat model could reach 64% (Figure 6b). However, when the data size continued to increase, the improvement in the accuracy of the flat model was not significant. Increasing

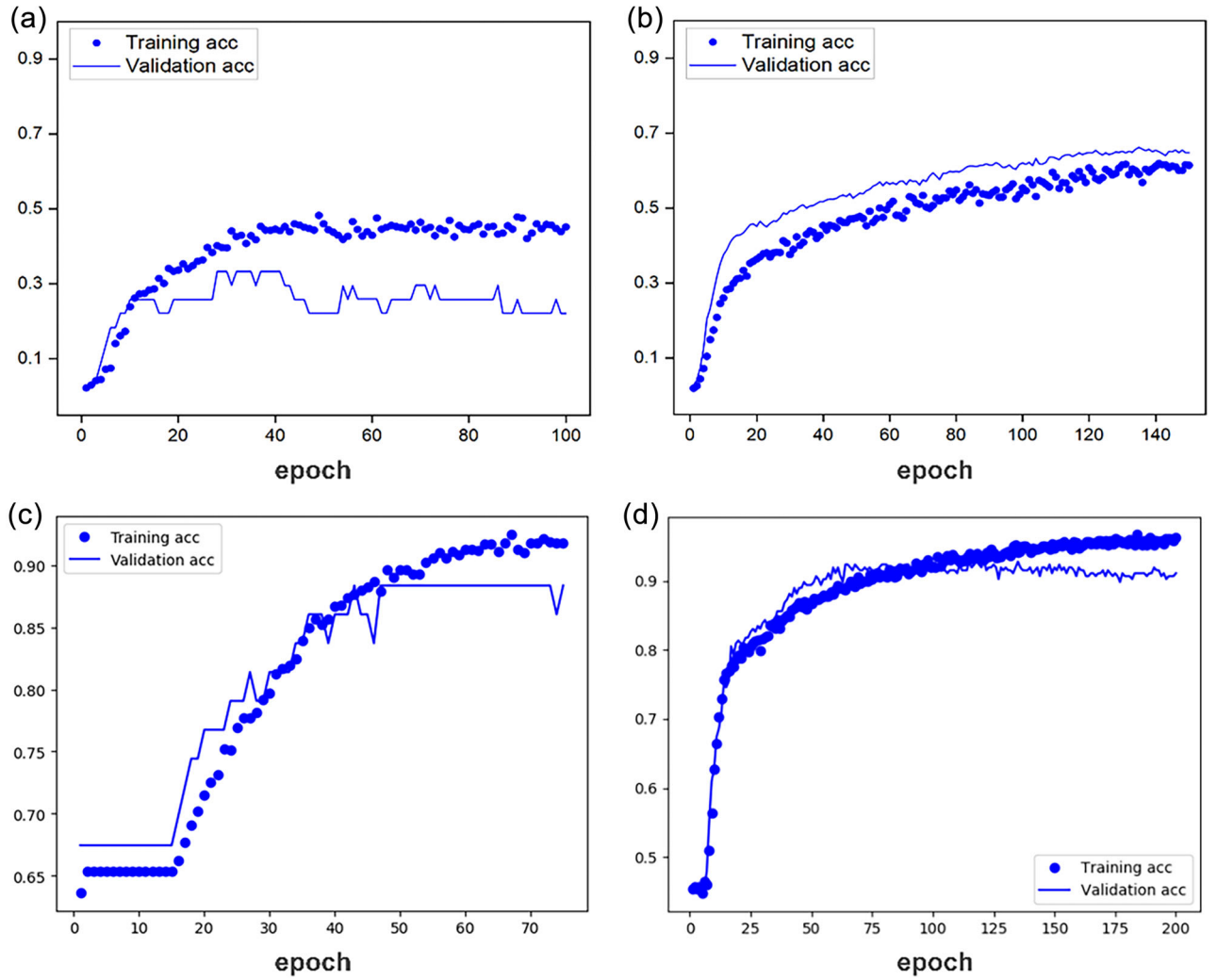


FIGURE 6 Training process of (a) flat network with limited data, (b) flat network with sufficient data, (c) sub-model of bearing, and (d) sub-model of substructure.

and adjusting the dataset size in the three-level sub-model can achieve an overall accuracy of 76.38%.

The results show that a flattened model requires more data to be trained, and its accuracy is poor when data are limited. However, when both flattened and multi-layer models are given sufficient training data, the accuracy gap between them becomes smaller but still exists.

Further observation of misclassified images in both models can provide insights into differences in accuracy. For instance, since the 24 defects are identified simultaneously, defects between different components tend to be confused with each other (Figure 7). However, through component-level filtering, the three-level model avoids some interference between defects. In addition, the flat model exacerbates the problem of sample imbalance. For example, in the dataset used herein, there are far fewer images of superstructures than on the bridge floor or substructure, which may be due to fewer components in superstructures. This type of sample imbalance may only manifest in the first-level model because the subsequent classification is irrelevant to this former data distribution. Since the Level 1 classification task is relatively simple, it is not much affected by the sample imbalance as well. However, in the flat model, the issue of sample imbalance is prominent since all categories are uniformly processed. Additionally, the hierarchical model facilitates the consistency in the feature sizes at the same level, which may explain the higher accuracy.

Although consistent VGGs are utilized for all tasks here, more targeted selection and optimization can be performed for different sub-tasks in the future. Additionally, a clear conclusion can be drawn from the above, that is, a three-level network is more advantageous for updating and incremental learning. For example, if a new type of deck defect needs to be added to the trained model, the flattened network may require overall retraining, while a

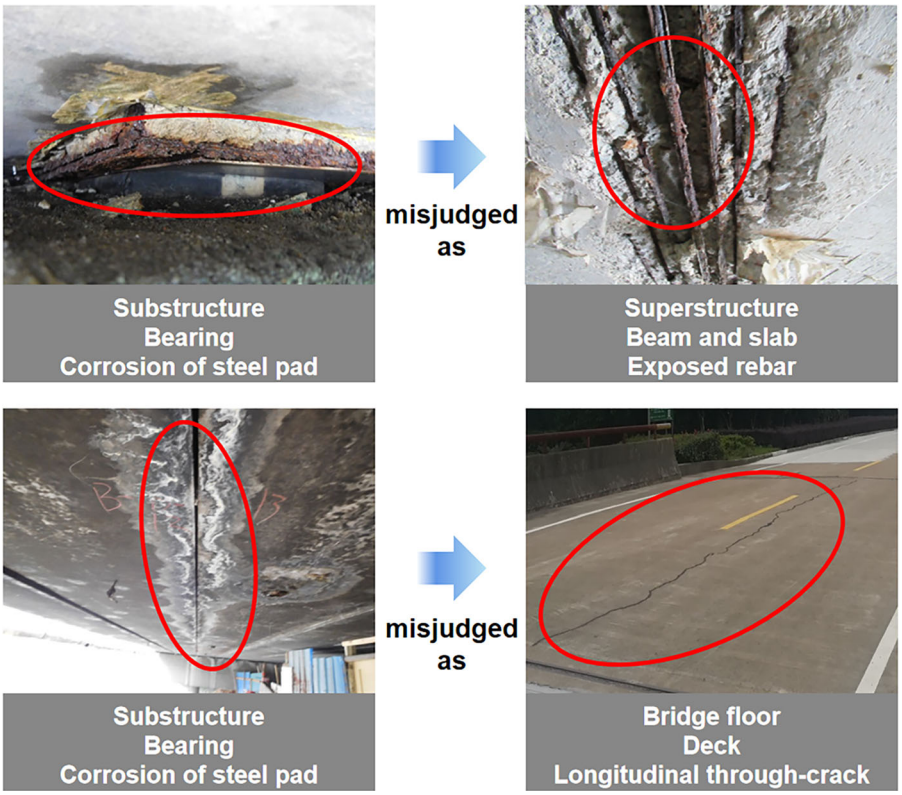


FIGURE 7 Defects between different components tend to be confused with each other in the flat network.

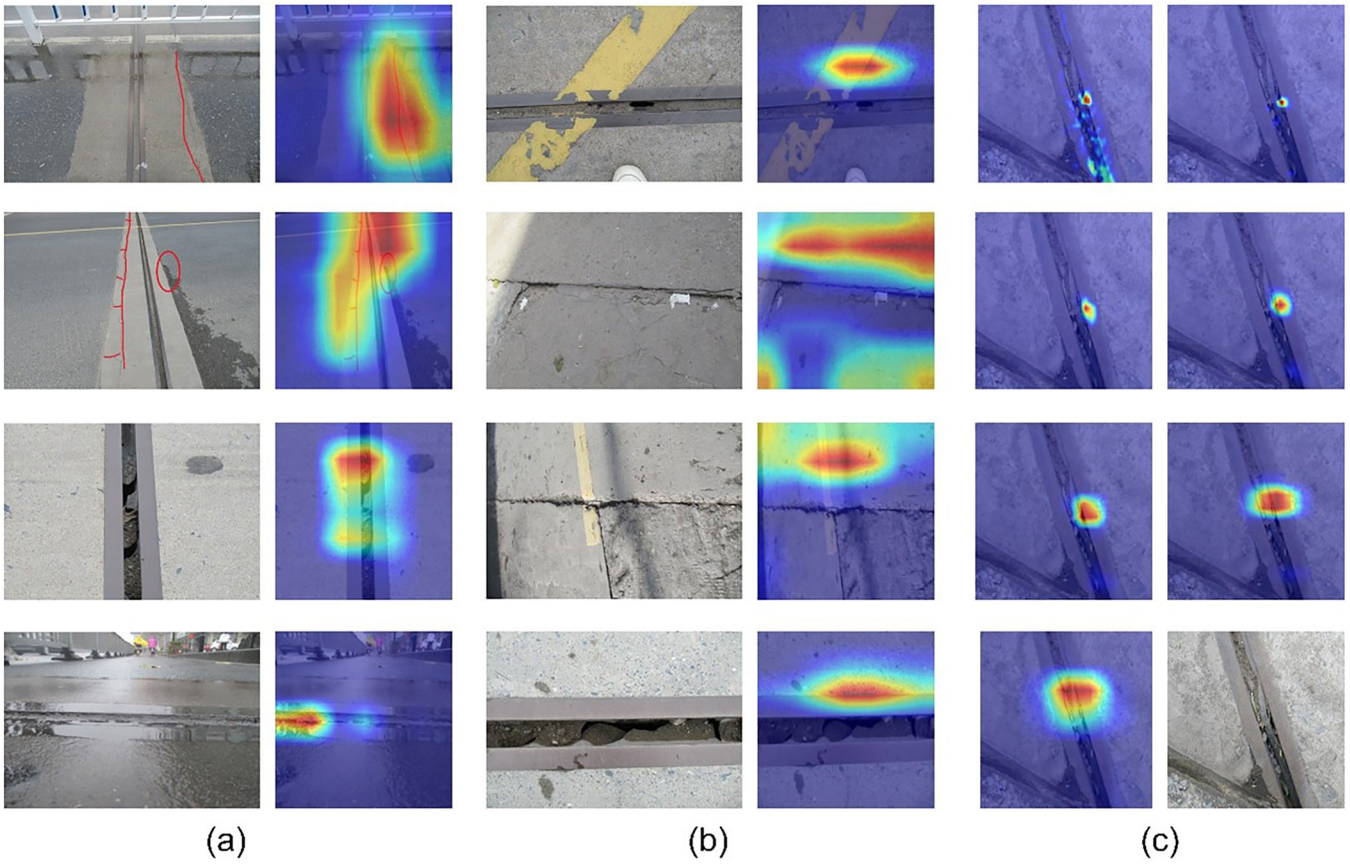


FIGURE 8 Grad-CAM visualizations for the "expansion joint defects" categories.



FIGURE 9 Examples of the classification results in the bridge structure level.

three-level network only needs to retrain a sub-network. In reality, engineering problems and data are constantly changing, and the model's scalability is also an important aspect.

4.3 | Interpretability

The proposed three-level model introduces the relationship between classes in the network. Compared to the flat model, the three-level model provides a visual way to understand the model's recognition process. Nonetheless, interpretability remains a challenge for DL-based approaches in general. Therefore, interpretable methods are utilized herein to further examine the neural network. Currently, Class Activation Mapping (CAM) proposed by B. Zhou et al. (2016) and Gradient-weighted CAM (Grad-CAM) (Xuhong Li et al., 2022) are two well-known techniques for visualizing and interpreting CNNs. More related work can be referred to Selvaraju et al. (2017) and Rosso et al. (2023).

By employing the Grad-CAM technique, a preliminary investigation into the network for expansion joint defects classification is conducted. After observing 250 visualized images, it is found that there are two typical types of activation regions in the correctly classified images by the network. The first type of activation regions overlap with the recognition object (Figure 8a), while the second type of activation regions is located at the boundary of the recognition object (Figure 8b). In addition, the outmost layer to the 20th layer from the end are activated for a specific image, and the representative ones are shown in Figure 8c. It can be seen that the activation range is large and intuitive in the earlier layers, then narrows, and gradually expands later. The alternation of activation area indicates that the network's attention becomes more focused and the features become more abstract from the bottom to the top layers.

5 | RESULTS AND DISCUSSIONS

The three-level classification of the BS information is obtained in the sequence of bridge structures, components, and defects and reflected on the given labels. First, the categories in the bridge structure level are identified and the images are classified into three types including bridge floor, substructure, and superstructure. The majority of the images can be correctly recognized (Figure 9) and the accuracy of the first-level classification reaches 95.99%.

Based on the first-level classification, the images labeled "bridge floor," "substructure," and "superstructure" are further classified into different types of bridge components. The images identified as the bridge floor in the first level are further classified into 11 categories of components and labeled as the handrail, the traffic sign, the drainage system, the deck, and so forth (Figure 10a). Results show that the second-level classification of the bridge floor is successfully realized with an average accuracy of 89.5%. At the same time, the images identified as superstructure are further classified into seven categories, including the girder bridge, arch bridge, and so forth. Herein, the drain holes are treated as part of the superstructure of the bridge because of their location. It can be seen in Figure 10b that the six types of urban bridge structures and the drain holes in the second level can be well identified with an average accuracy of 97.5%. Meanwhile, the images of the substructure are further divided into five categories of components such as wing wall, foundation, and so forth. The accuracy of the second-level classification of substructure components reaches 91.0%. Due to the different heights of the bridge piers, the pictures of substructures vary greatly in lighting conditions, which is one of the challenges for recognition. However, the components in the substructures are still successfully classified (Figure 10c), demonstrating the robustness of the model. The accuracy of components classification in the second level is in the range of 89.5%−97.5% (Table 5), in which 23 different kinds of components are considered.

Then, images labeled as 13 types of components in the second level are further classified into multiple types of defects (including the intact status). It can be seen that despite the slight inter-class differences, various types of

TABLE 5 The performance of the optimized model in second-level classification.

	Accuracy		Loss	
Parent directory	**Training**	**VLD**	**Training**	**VLD**
Bridge floor	0.9451	0.8953	0.4249	0.4952
Superstructure	0.9952	0.9748	0.0051	0.1147
Substructure	0.9502	0.9101	0.1003	0.2349
Average	0.9635	0.9267	0.1768	0.2816

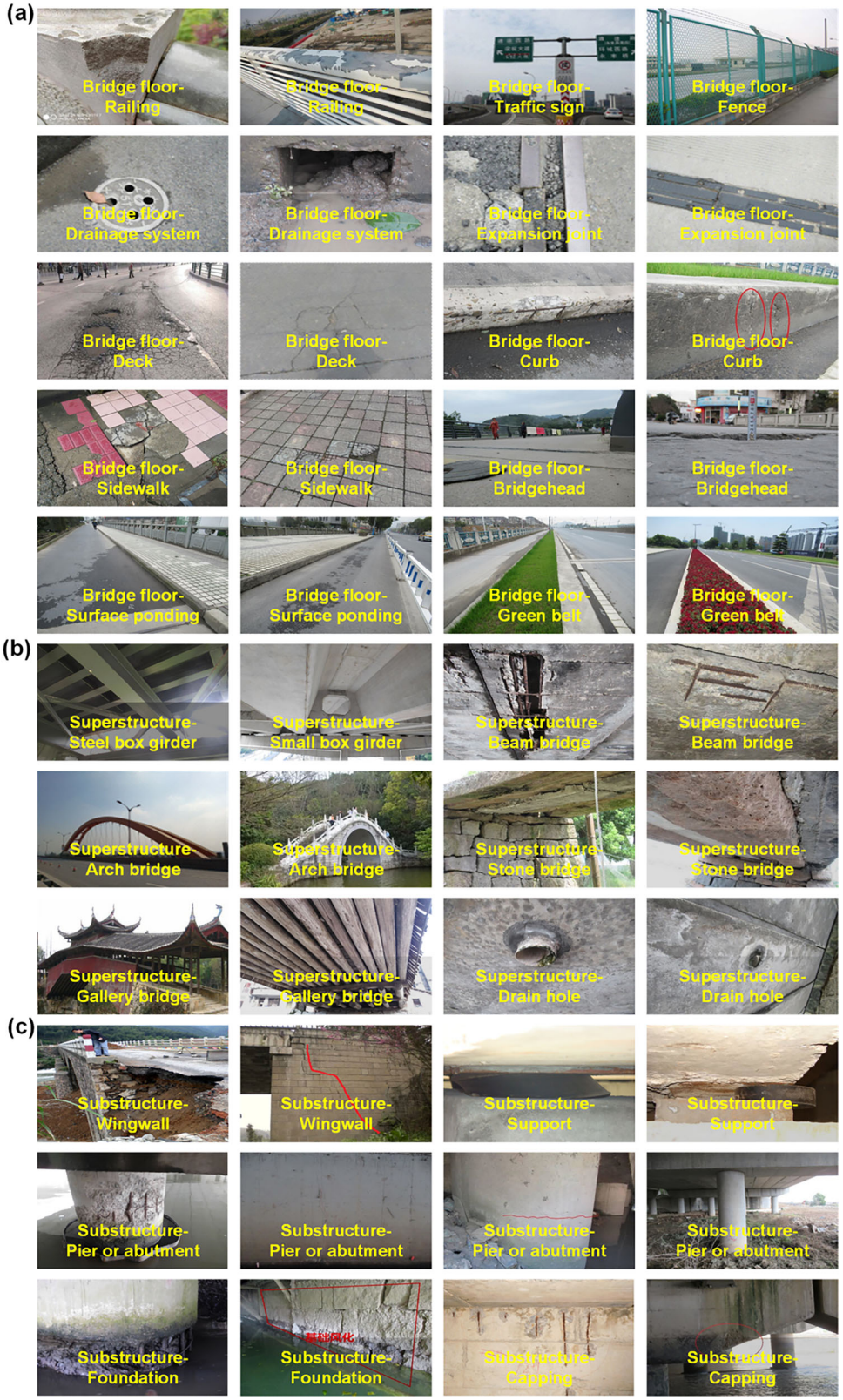


FIGURE 10 Examples of the classification results in the component level: components of (a) bridge floor, (b) superstructure, and (c) substructure.



FIGURE 11 Examples of the classification results in the defect level.

TABLE 6 The performance of optimized model for third-level classification.

	Accuracy (74 types)		Loss (74 types)		Accuracy (48 types)		Loss (48 types)	
Parent directory	**Training**	**Validation**	**Training**	**Validation**	**Training**	**Validation**	**Training**	**Validation**
Pier or abutment	0.9182	0.8837	0.3466	0.2149	0.9682	0.9461	0.1105	0.3215
Wing wall	1.0000	1.0000	0.0037	0.0039	0.9856	0.9895	0.0045	0.1021
Foundation	1.0000	1.0000	0.0000	0.0000	0.8749	0.8515	0.3427	0.4982
Handrail	0.8615	0.7834	0.3975	0.7031	0.8992	0.8848	0.2938	0.5687
Beam bridge-transverse	0.9824	0.9302	0.0468	0.4068	0.9945	0.9762	0.0297	0.3876
Beam bridge-beam and slab	0.8234	0.7822	0.5757	0.9555	0.9867	0.9541	0.1892	0.801
Expansion joint	0.7842	0.6014	0.5643	1.1896	0.9243	0.7864	0.2686	0.8231
Capping	0.8294	0.7708	0.466	0.7434	0.9015	0.8902	0.3546	0.7145
Deck	0.9149	0.6250	0.2652	1.2572	0.9246	0.8258	0.1856	0.9787
Sidewalk	0.8856	0.6956	0.4986	0.9912	0.8956	0.847	0.3874	0.7469
Bridgehead	0.6984	0.5995	0.6551	1.4321	0.7602	0.6214	0.8314	1.2021
Bearing	0.9673	0.8993	0.2123	0.4576	0.9765	0.9504	0.1878	0.4012
Drainage system	0.9317	0.9012	0.1989	0.3215	0.9875	0.9769	0.1497	0.2954
Average	0.8921	0.8056	0.3254	0.6674	0.9292	0.8846	0.2566	0.6032

defects could be successfully detected with our model (Figure 11). For the classification of 74 defects in the third level, the average accuracy is about 80.6%. In the labeling process of the 74 defects, some types of defects are found to be quite similar or usually occur simultaneously, and thus the strategy in the third-level classification has been modified. According to the new classification strategy, the same type of defects with different forms, such as through-transverse cracks and through-longitudinal cracks on the deck, are grouped with the same label. Meanwhile, the co-occurring defects are considered and labeled simultaneously like "exposed rebar and spalling." In the new classification strategy, the number of categories decreases from 74 to 48. As a result, the average accuracy rises to 88.5%, and the model is capable of detecting the images containing multiple defects simultaneously such as the image labeled as "bridge floor-sidewalk-missing and crack" in Figure 11. The specific accuracy and loss of defects located in the different components are depicted in Table 6.

To test the specific precision of the 48 kinds of defects and further validate the optimized model on new data, 100 new images for each category are collected as a test set. To evaluate the comprehensive classification ability of the model, only the images that are identified correctly in all the three levels are considered correct in this test. The average precision of the three-level classification is tested to be 80.4% (Table 7).

A total of 38,000 images are utilized to test the classification speed (Table 8), which are 17.2 and 5.76 FPS, respectively, in the single-level and three-level classification under Graphic Processing Unit (GPU), demonstrating the good efficiency of the proposed model.

6 | CONCLUSION AND DISCUSSION

In this paper, a hierarchical model is proposed to realize the classification and correlation of BS images at three levels: structural types, components (23 in total), and defects (75 in total). By doing so, this approach aims to provide a comprehensive understanding of bridge diagnosis. Accordingly, the following conclusions can be made:

1. A three-level classification model is proposed to classify the bridge structures, components, and defects jointly, so the impact of each defect on the bridge can be more clearly determined. The model consists of 17 trained sub-models, which are integrated to associate single-level detection. Compared to the traditional flat model, the proposed hierarchical network is advantageous in terms of accuracy, training simplicity, and scalability. Using the novel three-level classification algorithm, the overall training time of networks is shortened because the sub-models are trained parallelly. Meanwhile, some similar defects are triaged at the second level, thus avoiding interfering with each other and improving accuracy. The accuracy is also improved due to more specific and focused optimization of each sub-model.
2. To meet the training demand of the 100 classification targets in total and fully train the networks, more than 50,000 raw images are gathered from the on-site inspection. Six means of data augmentation are adopted to prevent overfitting. To broaden the datasets automatically, a labeling strategy is designed. In this process, the well-trained neural network is used in the classification and labeling process of new images instead of manually labeling work. In this way, the three tasks of the

TABLE 7 Test precision of the overall three-level classification.

Parent directory	Defect	Precision	Mean
Handrail	Loss or incomplete	1.0000	0.8629
	Looseness and dislocation	0.9534	
	Crack	0.7734	
	Exposed rebar	0.7246	
Deck	Wave or ruts	0.8308	0.7512
	Damaged	0.7143	
	Damaged and crack	0.8462	
	Crack	0.6135	
Sidewalk	Slack or deformed	0.8961	0.7779
	Missing	0.7500	
	Missing and crack	0.6781	
	Cracks	0.7874	
Bridgehead	Depression of abutment back	1.0000	0.3333
	Subsidence of bridgehead	0.0000	
	Intact	0.0000	
Expansion joint	Cracks	0.7197	0.8681
	Damaged	0.9327	
	Broken edges and cracks	0.8586	
	Intact	0.9615	
Beam bridge-Beam and slab	Spalling	0.9009	0.9146
	Cracks	0.8609	
	Cracks and spalling	0.9804	
	Exposed rebar	0.9583	
Beam bridge-Transverse connection	Damaged	0.9804	0.9251
	Looseness	0.8929	
	Crack	0.9615	
	Joint fracture	0.7813	
Capping	Intact	0.6369	0.7616
	Damaged	0.4050	
	Crack	0.8713	
	Damaged and crack	0.9615	
	Damaged and exposed rebar	0.9333	
Pier or abutment	Spalling	0.7384	0.8692
	Crack	1.0000	
Bearing	Intact	0.9091	0.9489
	Deformation of rubber bearing	0.9174	
	Damaged	0.9789	
	Steel bearing damage	0.9900	
Drainage system	Intact	0.8772	0.9509
	Missing or surface loss	0.9750	
	Missing and crack	0.9898	
	Surface ponding	0.9615	
Wing wall	Intact	1.0000	0.8395
	Spalling	0.5185	
	Crack	1.0000	
Foundation	Intact	1.0000	0.7747
	Scoured	0.5384	
	Damaged and crack	0.7857	
Average			0.8044

TABLE 8 The test speed of the classification.

Speed	GPU	Central Processing Unit (CPU)
Three levels classification	5.76 FPS	0.85 FPS
Single-level classification	17.2 FPS	2.55 FPS

dataset construction, model training, and classification are coupled together and updated jointly.

3. To improve the performance of the model, a series of training and optimization of hyperparameters is carried out to determine the suitable optimizer, learning rate decay strategy, epochs, and so forth. The classification of three types of structures, 23 components, and 74 defects with the well-trained and optimized model achieves accuracy of 96.0%, 92.3%, and 80.6%, respectively. Using the new classification strategy, the classification of 48 defects achieves 89% validated accuracy and a mean tested precision of 80.4% with the classification speed of 17.2 FPS. The outcomes prove the advantages of the proposed three-level model against ordinary flat models.

However, there are still some limitations in current research. In BS defect and damage detection, it is necessary to identify quantitative parameters such as the shape and area of the damage collection to further determine the severity of the damage. In the future, this framework should be further extended to tasks such as semantic segmentation, instance segmentation, and quantitative analysis. Although this study attempted to address the issue of multiple damage scenarios by setting categories for common combinations of damages, it cannot cover all possible situations. Therefore, in the future, further consideration is needed for multi-label problems. This study is still based on the classic VGG model to build sub-models. It is necessary to further investigate the advanced algorithms in the DL field, including the neural dynamic classification algorithm (Rafiei & Adeli, 2017a), dynamic ensemble learning algorithm (Alam et al., 2020),

self-supervised learning (Rafiei et al., 2022), and finite element (FE) machine (Pereira et al., 2020). The integration of these algorithms will certainly make this work more valuable.

ACKNOWLEDGMENTS

This work is supported by the National Natural Science Foundation of China (Grant Nos. 52122801, 11925206, 51978609, and U22A20254) and the Science Fund for Distinguished Young Scholars of Zhejiang Province (Grant Number LR20E080003).

DATA AVAILABILITY STATEMENT

Some or all data, models, or code generated or used during the study are available. The image data set is stored in the Figshare and can be visited through the following link: https://doi.org/10.6084/m9.figshare.24457549.

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How to cite this article: Zhang, H., Shen, Z., Lin, Z., Quan, L., & Sun, L. (2024). Deep learning-based automatic classification of three-level surface information in bridge inspection. *Computer-Aided Civil and Infrastructure Engineering*, *39,* 1431–1451. https://doi.org/10.1111/mice.13117

APPENDIX
THREE-LEVEL CLASSIFICATION ARCHITECTURE AND SAMPLE NUMBERS

Level	Parent directory	Category	Sample numbers
First level (structure)	–	Bridge floor	2963
		Superstructure	903
		Substructure	1839
Second level (component)	Substructure	Pier or abutment	3915
		Wing wall	1768
		Foundation	240
		Capping	3122
		Bearing	1546
	Superstructure	Beam bridge	4140
		Stone bridge	132
		Steel box girder	47
		Small box girder	70
		Gallery bridge	21
		Arch bridge	127
		Drain hole	357

(Continues)

Level	Parent directory	Category	Sample numbers
	Bridge floor	Sidewalk	1742
		Pavement	3606
		Drainage system	248
		Curb	526
		Railing	2349
		Expansion joint	1871
		Traffic sign	226
		Green belt	123
		Bridgehead	90
		Fence	16
		Surface ponding	55
Third level (defect)	Bearing	Concrete baseboard damage	113
		Corrosion of steel pad	388
		Steel bearing damage	40
		Deformation of rubber bearing	222
		Intact	163
		Bearing stability anomaly	14
		Damage of fixing bolt	9
	Foundation	Damage of concrete pile	1713
		Scoured	1606
		Hollowed foundation	10
		Intact	14
		Displacement of foundation	16
	Beam bridge-transverse connection	Spalling	343
		Net-shaped crack	1544
		Intact	148
		Joint fracture	399
	Beam bridge-beam and slab	Surface crack	452
		Concrete spalling	1084
		Structural crack	239
		Beam displacement	9
		Beam warping	6
		Crack seepage	248
		Intact	219
		Exposed rebar	234
	Handrail	Looseness and dislocation	325
		Exposed rebar	763
		Missing or incomplete	243
		Intact	36
	Drain system	Damaged waterproof	46
		Clogged	236
		Surface ponding	71
		Missing or surface loss	115
		Intact	36
	Pavement	Wave or ruts	53

(Continues)

Level	Parent directory	Category	Sample numbers
		Pit	87
		Pot hole	31
		Through-transverse crack	318
		Through-longitudinal crack	466
		Fragment or shatter	1142
		Net-shaped crack	530
		Intact	93
	Sidewalk	Missing	899
		Slack or displacement	178
		Polygonal or net-shaped crack	806
		Intact	96
	Expansion joint	Sediment blockage	200
		Steel material warping	15
		Height difference at the joint	18
		Broken edges	842
		Abnormal width of the joint	15
		Loose bolts	16
		Intact	19
		Damaged water stop belt	266
	Bridgehead	Depression of abutment back	10
		Intact	48
		Subsidence of bridgehead	67
	Wing wall	Spalling	1201
		Penetrating crack	367
		Lack of retaining function	15
		Intact	172
		Crack at junction	149
	Pier or abutment	Horizontal crack	98
		Longitudinal crack	62
		Spalling	302
		Frame-type node crack	15
		Intact	187
		Exposed rebar	200
		Leaning piers	10
	Capping	Surface cracks	39
		Chunk spalling	61
		Concrete spalling	1172
		Structural crack	451
		Crack seepage	323
		Exposed rebar	312
		Intact	183